Maka Nehith

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.32
Nano Junior College, Nallakunta, Hyderabad, Telangana	Class XII (Telangana Board of Intermediate Education)	2024	98.80%
Pallavi Model School, Boduppal, Ghatkesar, Hyderabad, Telangana	Class X (Central Board of Secondary Education)	2022	98.60%

SCHOLASTIC ACHIEVEMENTS

- Secured **Rank 1942** in the **IICPC CodeFest 2026 Global - Prelims Round** among 13,000+ participants
- Awarded **A+** grade in Operating Systems and **A** grade in Computer Architecture

RELEVANT COURSES

- **Systems:** Computer Architecture, Operating Systems, Database Management Systems, Software Development Fundamentals
- **Theoretical Computer Science:** Discrete Maths, Data Structures and Algorithms, Theory of Computation
- **Mathematics:** Probability and Random Variables, Calculus, Linear Algebra

PROJECTS

Implementation of New Features in xv6 | *Github Link*

- Extended the xv6 operating system by implementing **custom system calls**.
- Replaced the default **Round-Robin scheduler** with **System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ) scheduler** that prioritises interactive processes based on system call activity while maintaining fairness among processes.
- Extended the **virtual memory subsystem** by implementing **scheduler-aware page replacement** and **swap support**, enabling memory overcommitment while prioritising the working set of interactive processes.
- Implemented **FCFS** and **SSTF disk scheduling** policies, **disk-backed swap space**, and simulated **RAID 0, RAID 1 and RAID 5** for swap space management.

Speeding Up Integer Programs using FPU Parallelism | *Github link*

- Implemented **architecture-aware, code-level optimisations** to accelerate **integer-only C programs** by utilising idle Floating-Point Units(FPUs) for parallel execution.
- Improved the execution time for programs such as **matrix multiplication** and **minimum element search**, using **loop transformations** and **workload partitioning**.
- Evaluated the performance on **GEM5 RISC-V Out-of-Order Execution** simulator through execution time analysis.
- Achieved **up to 31.5% reduction in execution time** across evaluated integer benchmarks, with consistent improvements observed in compute-intensive workloads.

Bertrand's Paradox 3D simulation | *Github link*

- Developed a **3D Monte Carlo simulation** to visualise Bertrand's Paradox by comparing two random chord generation methods on a unit sphere.
- Implemented algorithms for uniform random sampling, vector-based geometric computations, and chord length

calculation within a unit sphere.
- Built an animated 3D visualisation with real-time probability estimation and statistical updates over configurable simulation iterations.

Arbitrary Precision Arithmetic Library in Java | *Github link*

- Implemented a Java library for arbitrary precision arithmetic using **object-oriented** programming concepts.
- Supports arithmetic operations on arbitrarily large integers and floating-point numbers **without loss of precision**.
- Packaged as a **JAR** file for easier integration into Java applications involving complex mathematical calculations.

SKILLS

- **Programming Languages:** C, C++, Python, Java, Shell Script, SQL
- **Libraries:** NumPy, Matplotlib, Pandas
- **Tools:** Git/Github, LATEX

EXTRACURRICULAR ACTIVITIES

- Represented the **school football team** in inter-school competitions
- Engage in recreational badminton and table tennis to promote physical fitness and foster social interaction